



उन्नत भारत अभियान Unnat Bharat Abhiyan



PROPOSAL (KRISHI VIMAN)

Proposal by:

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Subject expert group: Sustainable Agriculture System, Indian Agricultural Research Institute, Maharashtra

Title of the product / technology: Design and development of automated farming process through machine learning algorithms for khatmari Village.

Village where it is to be implemented: Khatmari Ta-Nagpur(Rural),Nagpur.

Why is this required: Under Unnat Bharat Abhiyan 2.0 we have successfully made farming easier by using machine learning in Vidarbha region.

Total cost of the product/technology: 40-45k

Commodities	Quantity	Amount
DJI pro drone with pre-installed mini camera (3.2-20MP)	1	20,000 - 30,000
Water cask (4-5L)	1	6,000 – 8,000
Sprinklers	4-5	1,000
Pipes	8-9	1,200 – 1,500
Fillers	6-7	1,000

Funds raised from gram panchayat or CSR or district/local administration?: No

Details of the funds raised from other agencies: Not applicable

Your role during the execution: Project developer with design and implementation within stipulated time and optimised budget, Supervisor.

Your role after installation: As a monitor, supervisor. Maintainer to control all parameters like batters, fillers, sprinklers & other basic development.

Brief description: Environment in Vidarbha region is not predictable for farming. Certainty of rainfall and appropriate environmental conditions are necessary for optimum plant growth and improved crop yields. In the conventional cultivation system, the farmer has to keep a watch on proper rain & light intensity, which is different for different crops. Dearth of water and uncertainty in rainfall in Vidarbha region are the prime reasons for low /no crop. This situation is persisting from year to year. The farmers takes loans to get a rid of this and could not pay in time. These are the reasons for their suicide under depression. If technology could assist in creating appropriate environmental conditions required for the crops, farmers will improve the crop yield and become economically strong. Hence proposed project will produce a ultimate solution for development & time saving for the villagers and definetly less emergent too. To make this drone our essentials/requirements are a DJI pro drone with pre installed mini camera in it the camera will be upto (3.2-20 MP) in that it determines from analog to digital with varies in the prices upto (20-30k). A simple tank will be connected to the drone upto 4-5 L Tank and its price max. goes upto (6-8k) determines by it's company & remaining that are sprinklers, pipes & fillers varies approx (3-4k). As a operator of this drone can easily covers upto 3 acers of land within just (15-20min). As well as the battery of the drone gets recharged within (30-40min). Hence the main advantage is that it reduces the wastage of the fertilizers/water, time & saves the farmers from some type of harmful pesticides too (low risk exposure). It is embaded with obstical radar which reduces the chances of accident & damage.

Impact of this work on learning of students/teachers: Involvement of institute in rural development. Students and teacher will learn how to transform technology for agricultural development in rural areas. Institute will highly associated with social responsibility. Students and Teachers will learn how to plan and execute technical solution with optimized cost. They will understand the development agenda within institutes of Higher Education and an institutional capacity and training relevant to national needs, especially those of rural India. Institute can prepare business plan after successful implementation. Beneficial for college research and development.

Duration of the work: 3-4 Months.

Final impact: Development and use of machine learning for rural development. All the functions to be performed by the Drone, Sprinkler and monitering of the camera. . Does not require any human intervention. Will strengthen the economy in rural areas which will ultimately help to develop economy of India. One time investment, all times gain.

